Strand 3: Geometry and Measurement

Sub-strand 2: Measurement

Lesson 4 (Requested topic): Area of Polygons — Square, Rectangle, Parallelogram, Triangle, Trapezium, Rhombus & Kite

1. Curriculum Standard and Overview

- Strand: 3 — Geometry and Measurement
- Sub-strand: 2 — Measurement
- Content Standard: B7.3.2.2 (area of plane shapes) carried into B8.3.2.1 measurement applications.
- Indicators: B7.3.2.2.1 (deduce the triangle area from a rectangle/parallelogram), B7.3.2.2.2 (determine the area of a triangle), B8.3.2.1.4 (apply it in real-life problems).
- Mock link: the 2010 Paper 2 trapezium item (area 31.5 cm^2, parallel sides 5.3 cm and 7.3 cm) is rebuilt in Example 5.

Prerequisites delivered: perimeter & area basics (this folder, Lesson 3) and triangle families incl. Pythagoras (Sub-strand 1, Lesson 4). You know what area means; today we collect formulas and, more importantly, learn which number is the height — the #1 exam trap.

2. Preamble — Base and Perpendicular Height

For every formula below, the two star numbers are:

- Base (b): any one side you choose to work from.
- Perpendicular height (h): the rapid distance from the base to the opposite corner/edge — always measured at right angles to the base, never along a slanted edge.

Trap: in a parallelogram or trapezium, the slanted edge is not the height. If a diagram gives you base 10, slant 6 and height 5, the area uses 10 × 5 — the 6 is a distractor (often placed there on purpose in BECE).

The master idea (indicator B7.3.2.2.1)

Any triangle is half of the parallelogram built around it. Draw a rectangle, draw one diagonal: two congruent triangles, each half the rectangle. That is the whole reason:

$$A_{\text{triangle}} = \frac{1}{2}bh, \qquad A_{\text{parallelogram}} = bh$$

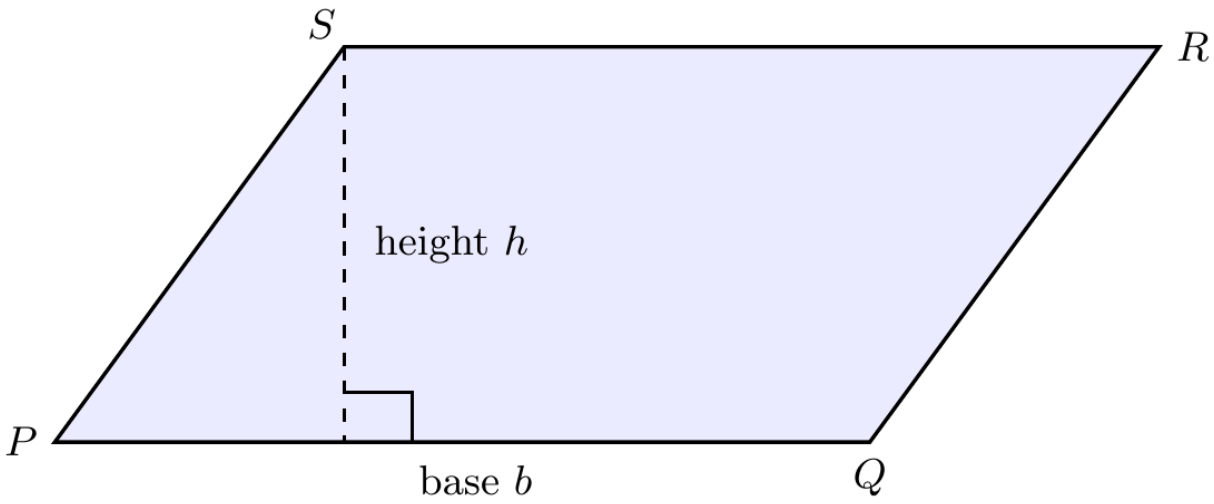
3. The Formula Table (learn by heart)

Shape	Area formula	Notes
Square (side s)	A = s^(2)	from Lesson 3
Rectangle	A = l × w	from Lesson 3
Parallelogram	A = b × h	h ⊥ to b
Triangle	A = ½ b h	any triangle, any chosen base
Right triangle	A = ½(leg_(1))(leg_(2))	the legs are base & height
Trapezium	A = ½(a + b)h	a, b are the two parallel sides
Rhombus / Kite	A = d_(1) × d_(2)/2	product of diagonals, halved

Reverse moves: h = 2A/b, b = 2A/h, h = 2A/(a+b).

4. Fifteen Fully Worked Examples

Example 1 — Parallelogram (own, uses figure)



A parallelogram has base 12 cm and perpendicular height 8 cm. Find its area.

A = bh = 12 × 8 = 96 cm^2.

Example 2 — The slant-edged distractor (own)

A parallelogram has base 10 cm, a slanted edge of 6 cm, and perpendicular height 5 cm. Find its area. Then find its perimeter if the opposite side is also 10 cm.

A = 10 × 5 = 50 cm^2. (6 is not the height!) P = 2(10 + 6) = 32 cm.

Example 3 — Triangle basic (own)

A triangle has base 14 cm and height 6 cm. Find its area.

A = ½(14)(6) = 42 cm^2.

Example 4 — Reverse (own)

The area of a triangle is 60 cm^2 and its base is 12 cm. Find its height.

h = 2A/b = 120/12 = 10 cm.

Example 5 — BECE 2010 Paper 2 (authentic, rebuilt cleanly)

The area of a trapezium is 31.5 cm^2. Its parallel sides are 5.3 cm and 7.3 cm. Find the perpendicular distance (h) between them.

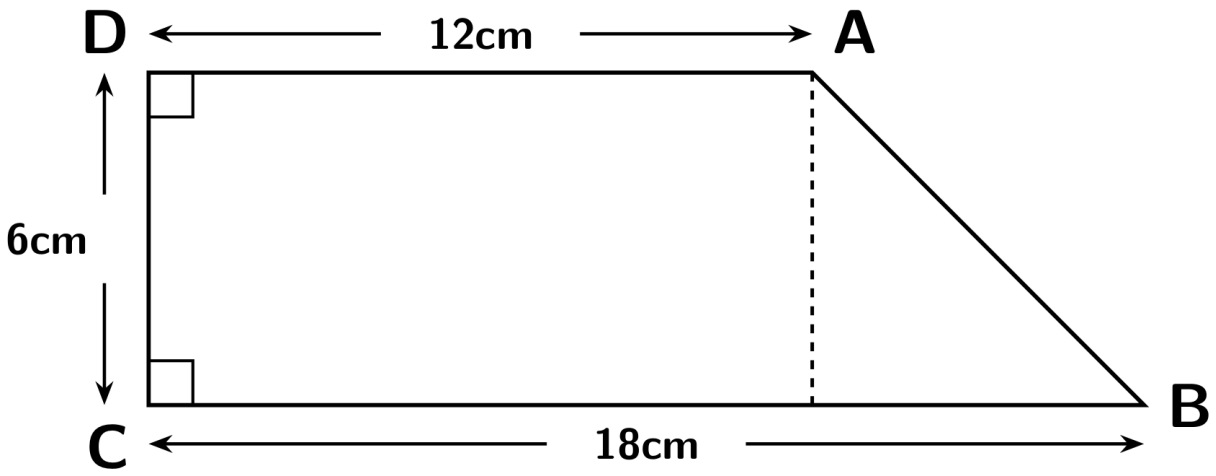
$$\frac{1}{2}(5.3+7.3)h = 31.5 \Rightarrow \frac{1}{2}(12.6)h = 31.5 \Rightarrow 6.3h = 31.5 \Rightarrow h = 5 \text{ cm}$$

Example 6 — BECE 1997 (authentic composite)

A figure is made of a rectangle 8 cm by 6 cm with a triangle on it, base 6 cm, height 3 cm. Find the total area.

Rectangle = 48 cm^2; triangle = ½(6)(3) = 9 cm^2; total = 57 cm^2.

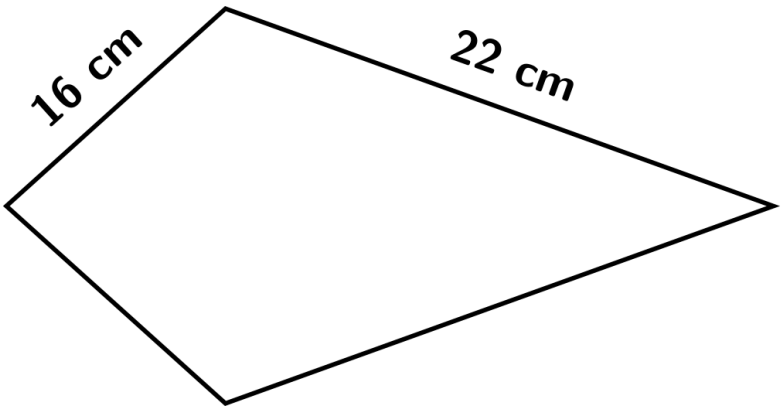
Example 7 — Trapezium basic (own, canonical figure)



A trapezium has parallel sides 10 cm and 6 cm, and height 5 cm. Find its area.

A = ½(10+6)(5) = ½(16)(5) = 40 cm^2.

Example 8 — Rhombus & kite (own, canonical figure)



NOT DRAWN TO SCALE

Find the area of (a) a rhombus with diagonals 10 cm and 24 cm; (b) a kite with diagonals 12 cm and 8 cm.

(a) A = 10 × 24/2 = 120 cm^2. (Bonus from triangles lesson: side = √(5^(2)+12^(2))=13, perimeter 52 cm.) (b) A = 12 × 8/2 = 48 cm^2.

Example 9 — BECE 1996 (authentic)

A square of side 6 cm has the same area as a rectangle of length 9 cm. Find the breadth of the rectangle.

Square = 36 cm^2. Breadth = 36 ÷ 9 = 4 cm.

Example 10 — Own (reverse parallelogram)

A parallelogram has area 240 cm^2 and base 20 cm. Find its perpendicular height, then its perimeter if one slanted pair of sides measures 13 cm.

h = 240 ÷ 20 = 12 cm. P = 2(20+13) = 66 cm.

Example 11 — Own (algebraic modelling, P2 style)

A triangle has base 2x cm and height x cm. Its area is 144 cm^2. Find x, the base, and the height.

½(2x)(x) = x^(2) = 144 → x = 12. Base = 24 cm, height = 12 cm.

Example 12 — Own (composite with grid logic)

A pentagon is a square of side 8 cm topped by a triangle of base 8 cm and height 6 cm. Find the total area.

Square = 64 cm^2; triangle = ½(8)(6) = 24 cm^2; total = 88 cm^2.

Example 13 — Real-life cost (own)

A farmer's plot is a trapezium: parallel sides 120 m and 80 m, distance between them 60 m. Weeding is charged at GH¢0.50 per m^2. Find the total cost.

A = ½(120+80)(60) = 6 000 m^2; cost = 6 000 × 0.50 = GHcent3 000.00.

Example 14 — Area from perimeter (own)

A square has perimeter 36 cm. Find its area. A rectangle with the same 36 cm perimeter is 11 cm long; find its area.

Square: s = 9, A = 81 cm^2. Rectangle: 2(11+w) = 36 → w = 7, A = 77 cm^2. (Again the square wins.)

Example 15 — In-scripted triangle inside a rectangle (own)

A rectangle is 20 cm by 14 cm. A triangle has one full side of the rectangle (20 cm) as base and the opposite side's midpoint as apex. Find the area of the triangle and of the leftover region.

Height of triangle = 14; area = ½(20)(14) = 140 cm^2. Leftover = 280 - 140 = 140 cm^2 (half the rectangle — diagonal-style split).

5. Exercises — 5 Sections × 5 Questions (No MCQ)

Show all working; state units.

Exercise 1 — Parallelograms

1. Find the area of a parallelogram with base 15 cm and height 9 cm.
2. A parallelogram has area 180 m^2 and height 12 m. Find the base.
3. A parallelogram's base and height are respectively 7 cm and (x - 3) cm; its area is 28 cm^2. Find x.
4. A parallelogram has base 11 cm, slant 6 cm, height 5 cm. Find (a) area, (b) perimeter.
5. A rhombus has side 9 cm and height 8 cm. Find its area. (A rhombus IS a parallelogram.)

Exercise 2 — Triangles

1. Triangle: base 18 cm, height 7 cm. Area?
2. Area 90 cm^2; base 15 cm. Height?
3. Right triangle with legs 8 cm and 11 cm. Area?
4. An isosceles triangle has equal sides 10 cm and base 12 cm. Use Pythagoras to find the height, then the area.
5. A triangle has area 42 cm^2 and height 12 cm. Find its base.

Exercise 3 — Trapezia, rhombi & kites

1. Parallel sides 13 cm and 7 cm; height 6 cm. Find the trapezium's area.
2. A trapezium's area is 84 cm^2 with parallel sides 10 cm and 14 cm. Find the distance between them.
3. A kite has diagonals 16 cm and 5 cm. Find its area.
4. A rhombus has diagonals 12 cm and 16 cm. Find (a) area, (b) one side, (c) perimeter.
5. A trapezium has parallel sides a cm and (a+4) cm, height 6 cm, area 66 cm^2. Find a.

Exercise 4 — Composite shapes

1. A figure is a rectangle 16 cm by 10 cm with a triangle attached on the 16 cm side, triangle height 6 cm. Find the total area.
2. A moon-shaped badge: a square of side 12 cm has a right triangle (legs 12 cm and 5 cm) cut from one corner. Find the remaining area.
3. An L-shape: big rectangle 14 cm by 9 cm minus a 6 cm by 4 cm corner. Find the area and perimeter.
4. A plot is a trapezium (parallel sides 40 m, 25 m, height 20 m). Find its area and the cost of fencing it at GH¢9.00 per metre if its non-parallel sides are 25 m each. (Perimeter = 40 + 25 + 25 + 25.)
5. A hexagon = rectangle 10 cm by 6 cm with one triangle (base 10, height 4) on top and an identical one below. Find the total area.

Exercise 5 — Spiral review

1. (Lesson 3) A rect floor 8 m by 5 m is tiled with 25 cm × 25 cm tiles. How many tiles?
2. (Triangles lesson) A triangle has sides 7, 24, 25. Name it and find its area (right angle between 7 and 24).
3. (Percent) A trapezium field of area 900 m^2 loses 15% to a pathway. Find the farmed area.
4. (Algebra) A parallelogram has base (2x) cm, height (x+3) cm and area 36 cm^2. Form the equation, solve for x, and find the base and height.
5. (Congruence/similarity) Two similar triangles have areas 50 cm^2 and 200 cm^2. What is the ratio of (a) their sides, (b) their areas? (Think: similarity from your earlier lesson, scale factor k on lengths → area factor k^(2).)

6. Full Worked Solutions — Exercises

Exercise 1

1. 15 × 9 = 135 cm^2.
2. b = 180 ÷ 12 = 15 m.
3. 7(x-3) = 28 → x - 3 = 4 → x = 7.
4. (a) 11 × 5 = 55 cm^2. (b) 2(11+6) = 34 cm.
5. Rhombus = parallelogram: A = 9 × 8 = 72 cm^2.

Exercise 2

1. ½(18)(7) = 63 cm^2.
2. h = 2 × 90/15 = 12 cm.
3. ½(8)(11) = 44 cm^2.
4. Height = √(10^(2) - 6^(2)) = √(64) = 8 cm; A = ½(12)(8) = 48 cm^2.
5. b = 2 × 42/12 = 7 cm.

Exercise 3

1. ½(13+7)(6) = 60 cm^2.
2. ½(24)h = 84 → h = 7 cm.
3. 16 × 5/2 = 40 cm^2.
4. (a) 12×16/2 = 96 cm^2. (b) side = √(6^(2) + 8^(2)) = 10 cm. (c) P = 40 cm.
5. ½(a + a + 4)(6) = 66 → 3(2a+4) = 66 → 2a + 4 = 22 → a = 9.

Exercise 4

1. 160 + ½(16)(6) = 160 + 48 = 208 cm^2.
2. 144 - ½(12)(5) = 144 - 30 = 114 cm^2.
3. Area = 14×9 - 6×4 = 126 - 24 = 102 cm^2; perimeter (walk): 14 + 9 + 6 + 4 + 6 + 5 = 44 cm — check: full rect 2(14+9) = 46 ≠ 44. Walk honestly with the notch 6 wide, 4 deep at a top corner: bottom 14? For a

top-right notch: 14 + 5 + 6 + 4 + 8 + 9... Set vertices:
(0,0)→(14,0)→(14,9)→(8,9)→(8,5)→(2,5)→(2,9)→(0,9)→(0,0) — that's a U! Simpler single notch:
(0,0)→(14,0)→(14,9-4=5)→(14-6=8,5)→(8,9)→(0,9)→(0,0): sides 14+5+6+4+8+9 = 46 cm = 2(14+9)
(staircase notch preserves total).
4. Area = ½(65)(20) = 650 m^2. Perimeter = 40 + 25 + 25 + 25 = 115 m; cost = 115 × 9 = GHcent1,035.00.
5. 60 + ½(10)(4) × 2 = 60 + 40 = 100 cm^2.

Exercise 5

1. Floor = 40 m^2 = 400 000 cm^2; tile = 625 cm^2; tiles = 640.
2. 7^(2) + 24^(2) = 625 = 25^(2) → right-angled scalene; A = ½(7)(24) = 84 $units^2$.
3. Farmed = 900 × 0.85 = 765 m^2.
4. (2x)(x+3) = 36 → x^(2) + 3x - 18 = 0 → (x+6)(x-3) = 0 → x = 3 (reject −6). Base = 6 cm, height = 6 cm.
Check: 6 × 6 = 36 . (Exam habit: check your roots are sensible — lengths cannot be negative.)
5. (a) Area ratio = 200÷50 = 4, so side ratio = √(4) = 2 : 1. (b) 4 : 1.

7. Memory Card

- Parallelogram = bh; triangle = ½bh; trapezium = ½(a+b)h; rhombus/kite = d_(1) d_(2)/2.
- Height is never the slanted edge.
- Reverse: h = 2A/b.
- Similar shapes: sides in ratio k → areas in ratio k^(2).

Next lesson: Parts of a circle — the last shape family, ruler-free: centre, radius, diameter, chord, arc, sector, segment, tangent, and the π formulas that measure them all.